RecoverIQ

QUASAR 5.0 · PS-01 FINTECH · AI-POWERED DEBT COLLECTIONS

One case. One trail. Full visibility.

RecoverIQ — an explainable, offline-first, compliance-by-design debt collections platform for banks and NBFCs

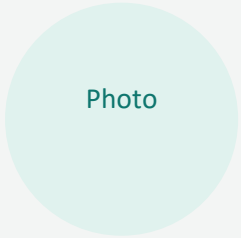
18 / 18
essential requirements built

0
payments marked paid without the gateway

34
automated browser checks passing

TEAM

The team behind RecoverIQ

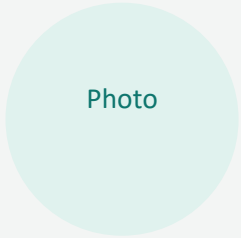


Member name

College · Year

Backend & Data

FastAPI, state machine, payments & webhooks, audit chain, offline sync

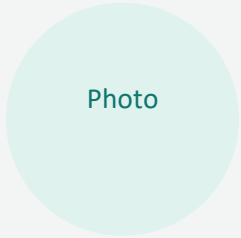


Member name

College · Year

Frontend & Mobile

Manager console, agent PWA, maps, borrower pay page

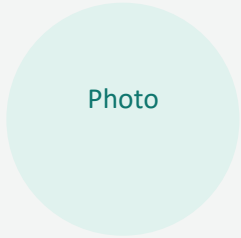


Member name

College · Year

AI & Integrations

Scorecard + self-cure model, multilingual sentiment, contact policy



Member name

College · Year

Product & Pitch

Demo story, compliance content, testing, decks and video

THE PROBLEM

Collections still run on spreadsheets, phone calls and trust

~40%

of fresh NPAs in FY24 came from retail non-home loans, much of it through NBFC and fintech digital lending.

Source: RBI Financial Stability Report, June 2024 (as summarised in the plan's sources)

1. **Every account treated alike**

   A ₹4,000 account 5 days late gets the same visit as a ₹4 lakh account 85 days late.

2. **No field visibility**

   Managers can't see who visited whom, or whether the visit happened at all.

3. **Promises go missing**

   "I'll pay on the 10th" lives in a notebook. Broken promises go unnoticed for weeks.

4. **Payments posted by hand**

   Double-posting, and cases marked paid when the money never arrived.

5. **Compliance hard to prove**

   RBI expects contact only 8 AM–7 PM and fair conduct, with no reliable log to show it.

ROOT CAUSE

Fragmented tools mean there is no single case record

Today

- Excel lists split by whoever is free
- Phone and WhatsApp updates at day's end
- Payment status typed in by staff
- No record of contact times
- Same treatment for every borrower

With RecoverIQ

- One case per account, ranked by explainable priority
- GPS-stamped visits that sync even from offline phones
- PAID only on a signed gateway webhook
- Contact Policy Engine blocks 8 PM calls and logs why
- Field visit, call or digital nudge chosen per case

BENEFICIARIES

Everyone in the collection chain gets something concrete

1

Lenders

Banks, NBFCs, MFIs: faster recovery, lower cost to collect, provable compliance

2

Field agents

A ranked worklist that works offline and credit only for verified work

3

Managers

Live field map, allocation, escalations, one consistent record

4

Borrowers

Contact only in fair hours, in Hindi, Marathi or English; self-pay link, instant receipt

5

Regulators & auditors

Tamper-evident trail of every contact, reveal and payment

Built for Bharat-scale field operations: patchy networks, many languages, strict fair-practice rules.

OUR SOLUTION

The lifecycle of one case, end to end

1

Intake

CSV / Excel, one case per account, row-level errors

2

Score

0–100 risk with reasons + self-cure %

3

Route

Field · Tele · Digital by versioned rules

4

Work

Allocation, offline agent app, nudges

5

Pay

Link / UPI QR → gateway → signed webhook

6

Learn

Outcomes retrain the model; manager approves

Every step writes to one shared case record and a hash-chained audit trail

Same case open on three screens updates together over WebSocket · status changes only through the backend state machine · PAID and CLOSED only from the payment service

WHAT'S NEW

Seven ideas that make RecoverIQ different

1

Explainable dual-layer scoring

Rule scorecard shows every point; a logistic self-cure model adds exact per-feature attributions.

2

Payment truth from the gateway

HMAC-verified webhooks, unique payment IDs, reconciliation for lost webhooks.

3

Offline-first agent app

IndexedDB outbox with client UUIDs, idempotent sync, conflicts ruled out by design.

4

Compliance as code

Every SMS, WhatsApp, email and call passes the 8 AM–7 PM window, caps and opt-outs.

5

Tamper-evident audit

SHA-256 hash chain; the database refuses UPDATE and DELETE on the log.

6

Multilingual sentiment map

Hindi, Marathi, English notes → mood and reason → state, city, borrower.

7

Human-approved learning loop

Retrain → shadow comparison → manager approves → rollback any time.

Portfolio health at a glance, live



Six KPIs

Overdue book, collected, collection rate, cost per ₹100, digital self-cure, promises kept

Ageing & channel mix

DPD buckets SMA-0 → NPA and Field / Tele / Digital split

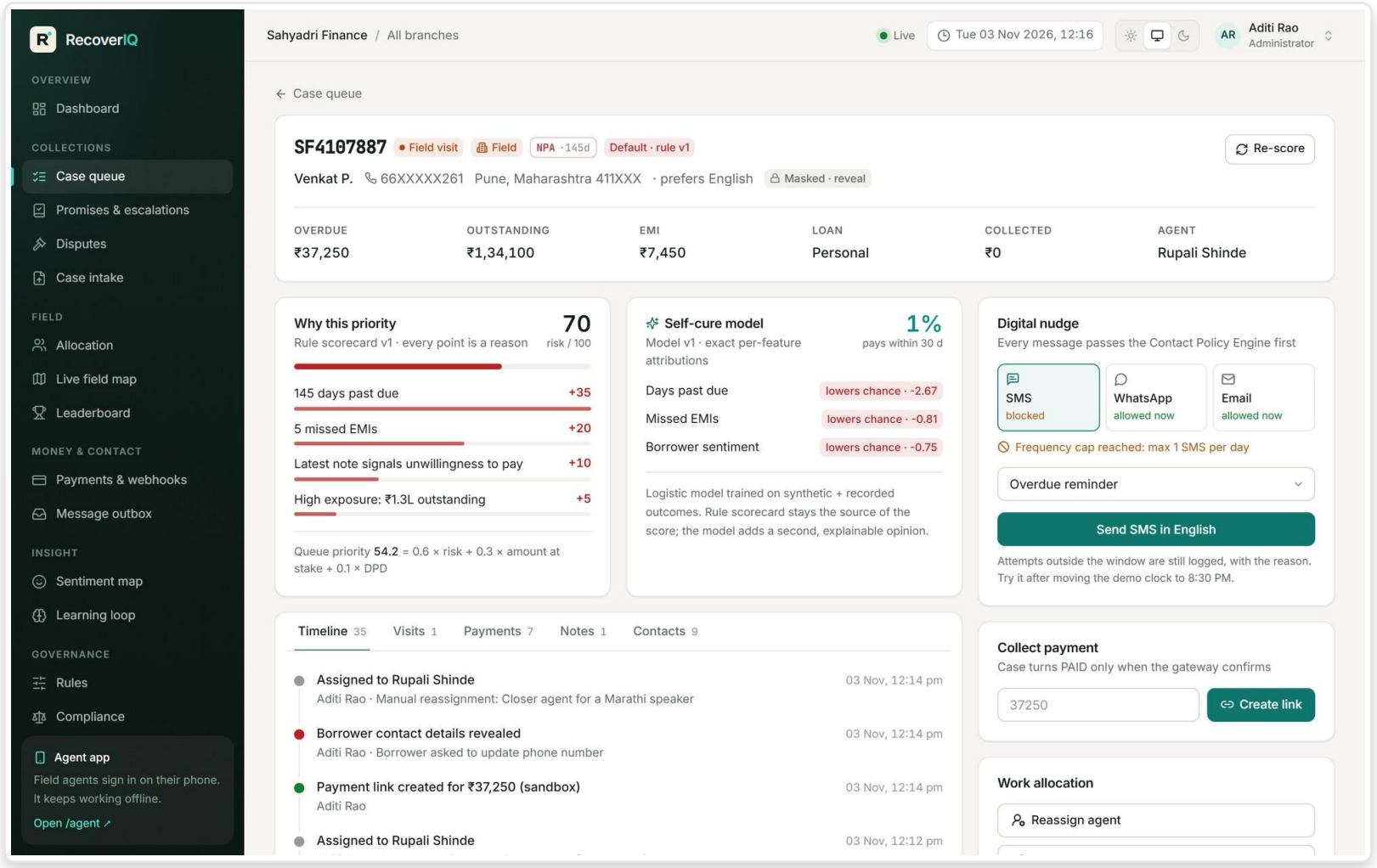
Live activity

Check-ins, payments and broken promises arrive in real time

Scoped by role

Branch, region or all-India; PII masked for managers

Every score comes with its reasons



Scorecard reasons

Each point added or removed is shown, with the rule version used

Self-cure model

Probability of paying in 30 days with its top three drivers

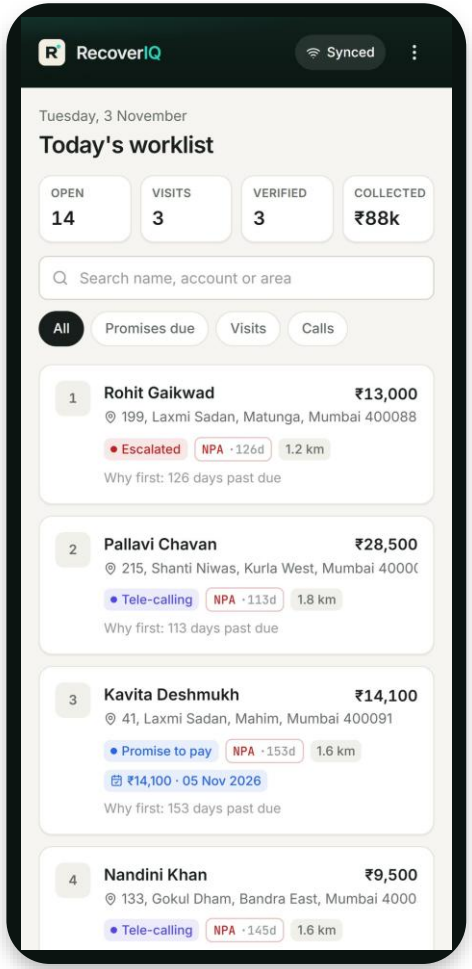
One timeline

Intake, routing, visits, notes, nudges, payments: one record

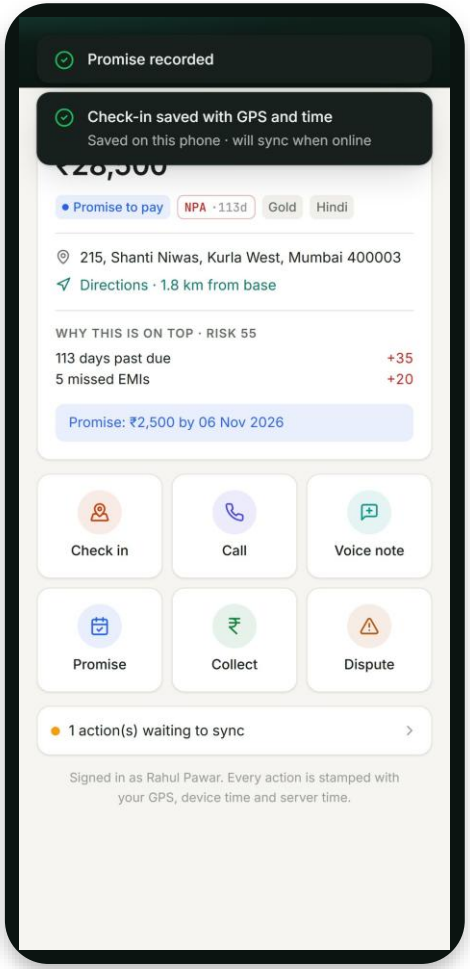
Actions with reasons

Reassign, override channel, reveal PII: all audited

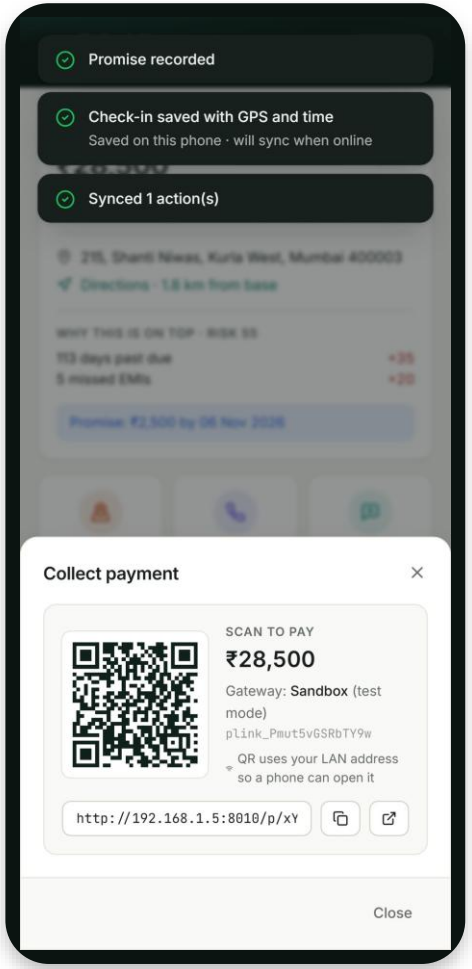
A worklist that keeps working with no network



Ranked worklist



Offline: actions queued



UPI QR to collect

GPS + time on every check-in

Within 300 m = verified; far, low-accuracy or impossible travel speed = flagged

Outbox with client UUIDs

Visits, notes, promises sync later; the server applies each exactly once

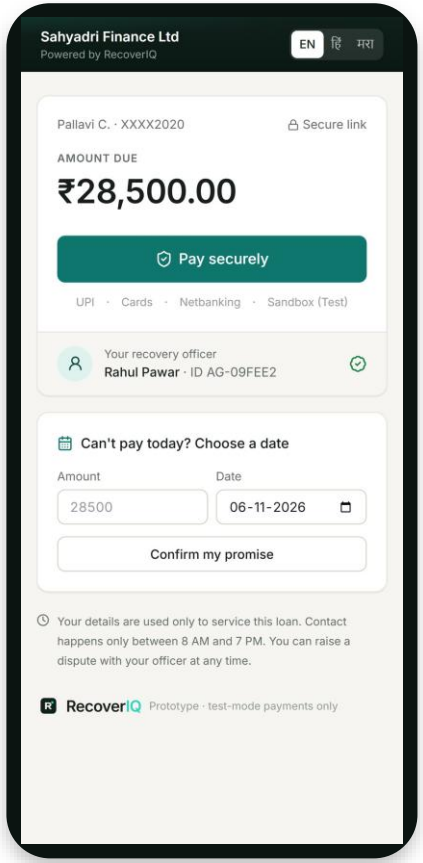
Voice notes

Record and transcribe in Hindi, Marathi or English; edit before saving

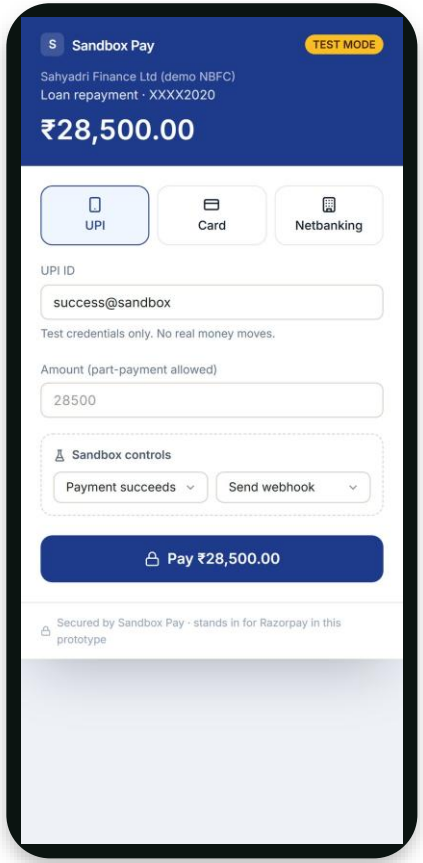
Installable PWA

Add to home screen on any Android or iPhone

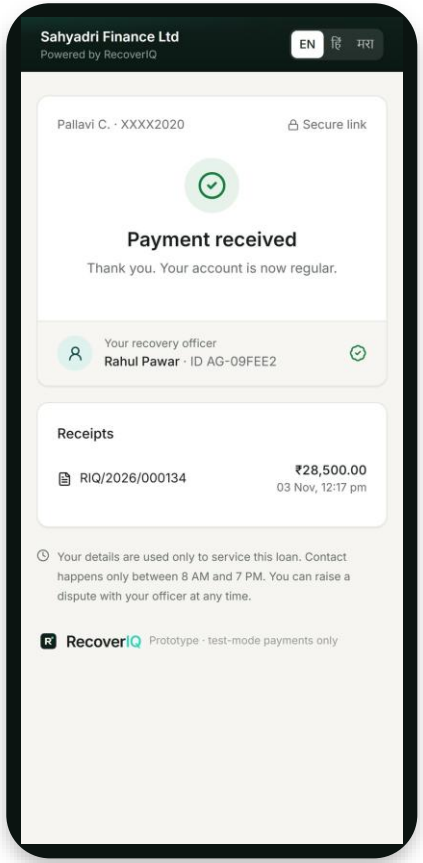
A case turns PAID only when the gateway says so



1 · Borrower pay page



2 · Gateway checkout



3 · Confirmed + receipt

Proven in the demo

Signed webhook

HMAC-SHA256 verified before anything is posted

Replay → duplicate_ignored

Unique gateway payment ID: no double posting

Forged event → 401

Wrong signature is rejected and logged

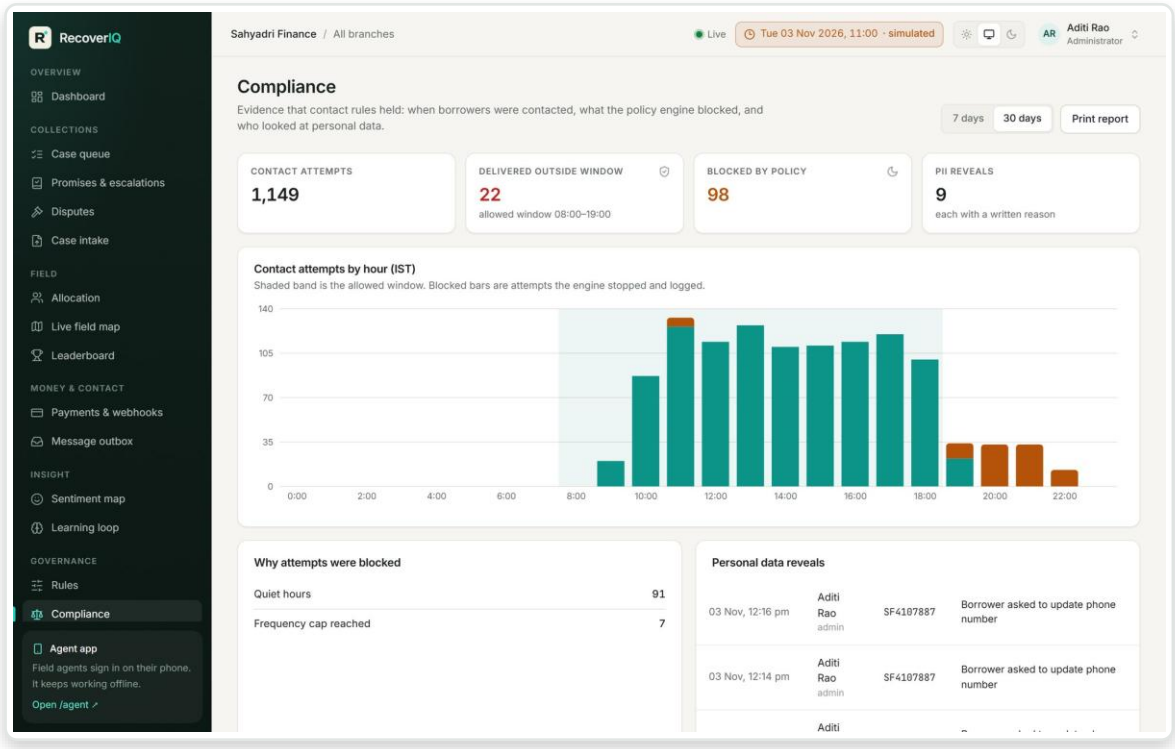
Lost webhook → reconciliation

Job asks the gateway and posts what's missing

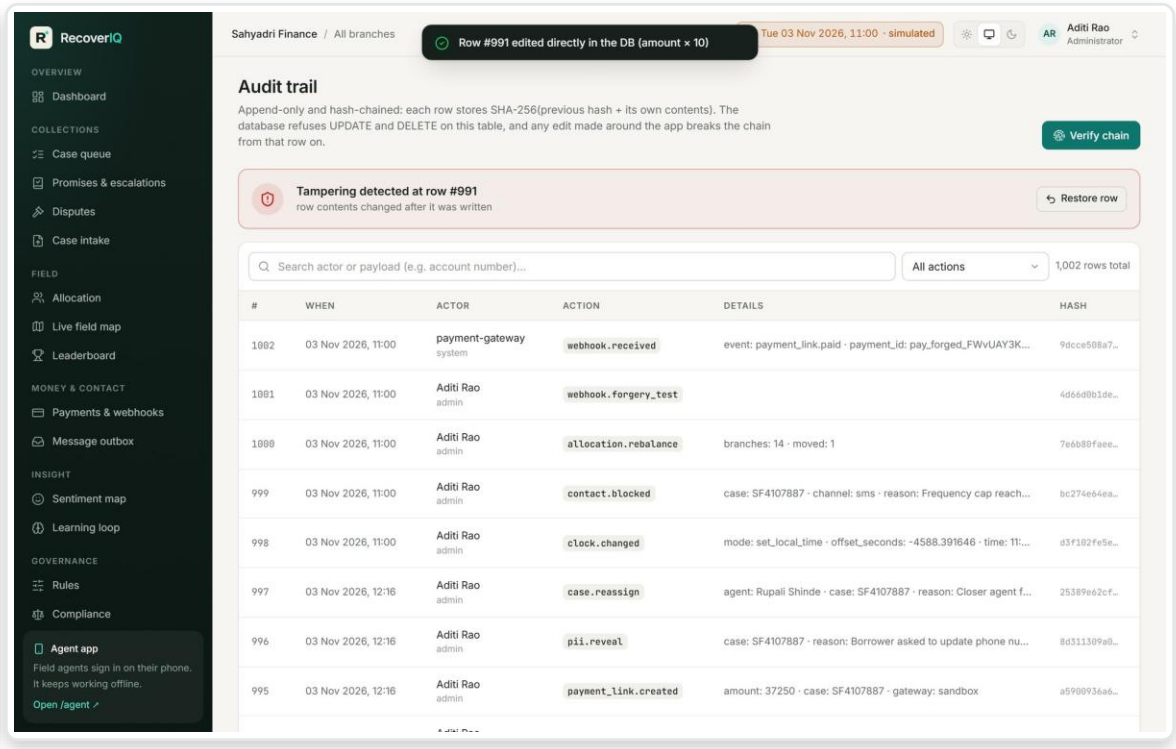
Instant receipt

PDF receipt for the borrower and the branch

Fair-practice rules enforced in code, and provable

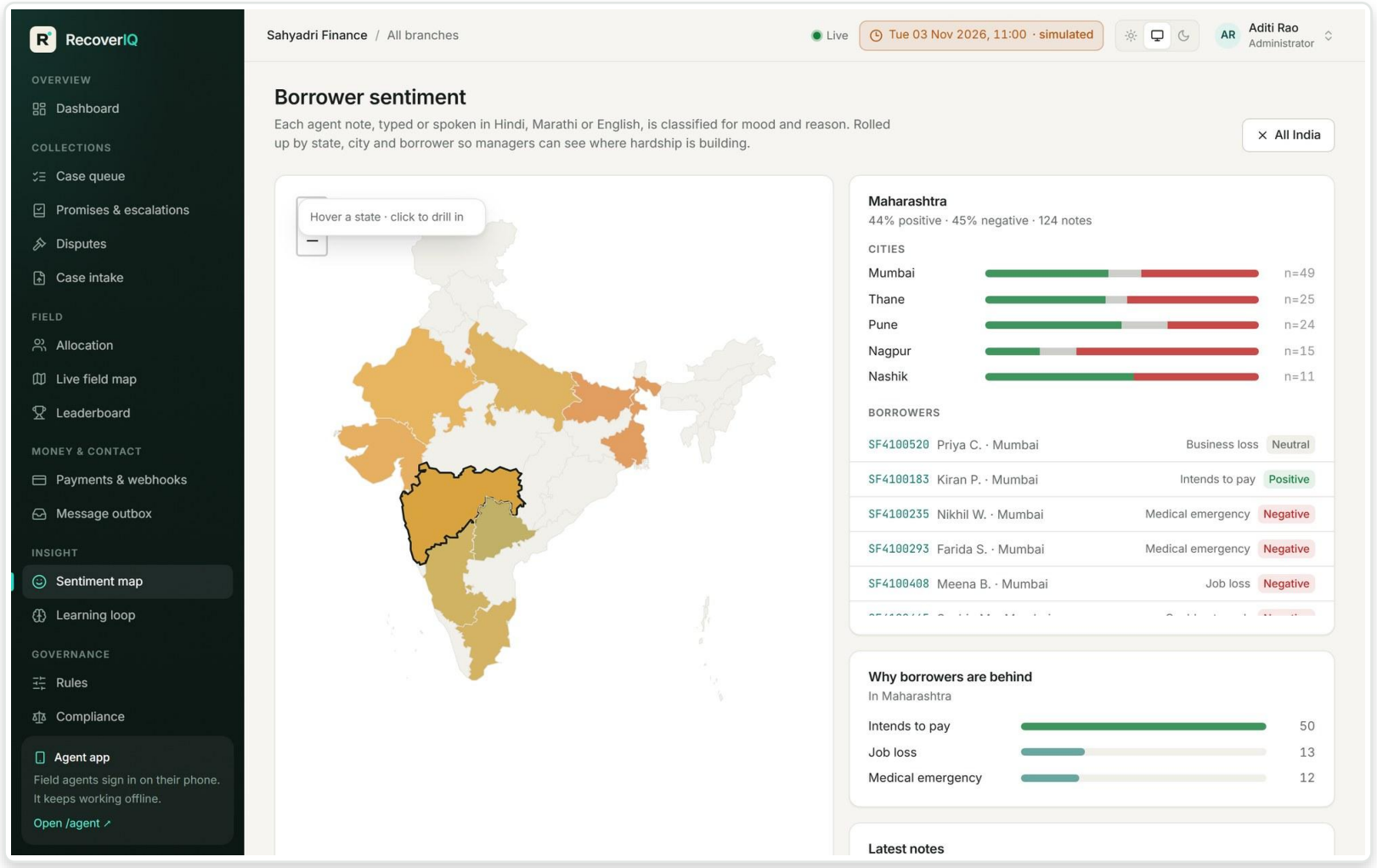


Contact attempts by hour: nothing delivered outside 8 AM–7 PM; blocked attempts logged with the reason.



Audit trail: an edit made directly in the database breaks the hash chain at that exact row.

Hardship mapped from what borrowers actually say



Three languages

Hindi, Marathi, English and Hinglish, typed or spoken

Mood + reason

Job loss, medical, business loss, dispute, unwilling, intends to pay

State → city → borrower

India map with drill-down and recent notes

Feeds the score

An 'unwilling to pay' note adds risk points, visibly

REQUIREMENT COVERAGE

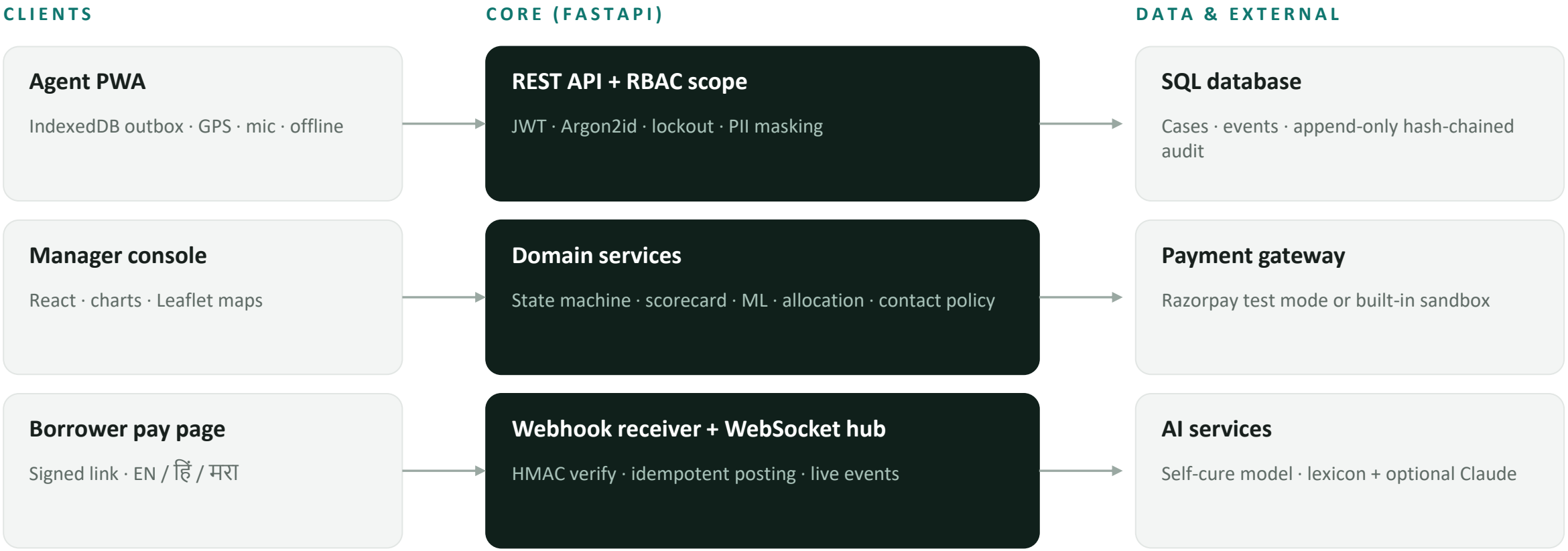
18 of 18 essential requirements, each with a live proof

Requirement	Proof in the demo
1 Case intake	Re-upload same CSV → 0 new cases
2 Risk scoring	Reasons panel on every case
3 Channel decision	Change threshold → cases re-route
4 Allocation	Rebalance evens workload bars
5 Agent app	Ranked worklist on a phone
6 Visit tracking	GPS check-in → verified / flagged
7 Digital nudges	8:30 PM nudge blocked + logged
8 Payment collection	UPI QR, pay link, PDF receipt
9 Payment confirmation	Replay webhook → no double post

Requirement	Proof in the demo
10 PTP & escalation	+3 days → promise breaks → escalates
11 Dashboards	Live KPIs, ageing, field map
12 Offline	No network → check-in → sync
13 Permissions	Exec sees 98XXXXX321; reveal needs reason
14 Audit & compliance	Verify chain; tamper → detected
15 Default rule	Edit expression → reclassify
16 Sentiment view	India map, state → city → borrower
17 Note capture	Hindi / Marathi voice → text
18 Single case record	Same case updates on 3 screens

ARCHITECTURE

Stateless API, one source of truth, payment truth from the gateway



Background jobs: promise-break scheduler (every minute) · gateway reconciliation (every 10 minutes) · nightly retrain → shadow → approve

DATA MODEL

Three constraints carry most of the correctness

UNIQUE(account_no)

One case per loan account. Re-uploading a file updates, never duplicates.

UNIQUE(gateway_payment_id)

A replayed or duplicate webhook can't post money twice.

UNIQUE(client_id)

Offline actions carry device-made UUIDs, so a resent outbox applies once.

16 tables

cases	borrowers (encrypted PII)
case_events	assignments
visits	contact_attempts
promises	payment_links
payments	webhook_events
notes	disputes
rule_versions	model_versions
audit_log (hash-chained)	users · branches · regions

Chosen so the demo works on stage, every time

Frontend	React 18 · TypeScript · Vite · Tailwind · Recharts · Leaflet · Dexie (IndexedDB) · PWA
Backend	Python · FastAPI · SQLAlchemy 2 · Pydantic v2 · WebSocket
Data	SQLite (WAL) for the laptop demo; Postgres-ready via DATABASE_URL
AI / ML	scikit-learn logistic model, exact attributions · multilingual lexicon · optional Claude
Security	JWT + refresh · Argon2id · AES-GCM PII · HMAC webhooks · RBAC scopes
Payments	Razorpay Payment Links (test) or built-in sandbox gateway

27

unit, API & integration tests passing

34

browser steps passing, 0 console errors

5 roles × 31

endpoints smoke-tested, 0 server errors

740

synthetic accounts across 14 branches

EXPECTED IMPACT

Targets for a one-branch pilot, not claims

20–30%

of low-risk accounts self-cure digitally, no visit needed

↓ cost

to collect: visits go to the accounts where they matter

0

contacts outside 8 AM–7 PM, with the log to prove it

Same day

visibility of every visit, promise and payment

Business model

Customers

NBFCs, small finance banks, MFIs, co-op banks, fintech lenders, agencies

Pricing

Per active delinquent case per month, or per agent seat; enterprise VPC tier

Go-to-market

Pilot one branch → measure cost per recovery & self-cure → expand by region

SCALABILITY & ROADMAP

From one branch to multi-region, and on to the finale

Pilot

1 branch · 20 agents · ~5k cases

1 API · managed Postgres

Region

10 branches · 300 agents · ~100k cases

API autoscale · worker pools · read replica

Multi-region

5k agents · ~5M cases

Partition by tenant + region · OLAP for dashboards

Done

Working prototype: all 18 requirements, tests, demo data

Round 0 · Dec

This deck + 90-second demo video

Round 1 · Jan

5-minute engineering pitch + live link

Finale · 23–24 Jan

Judge feedback → build → 'You said, we built'

Next

Voice-bot reminders · route optimiser · Account Aggregator affordability

LIVE DEMO

Seven minutes, eight moments judges can verify

1. **Intake**
   Upload sample CSV twice: 0 duplicates

2. **Score & route**
   Open a case: reasons; move a threshold: cases re-route

3. **Allocate**
   Rebalance: workload bars even out

4. **Agent phone**
   Offline check-in + Marathi note → reconnect → synced

5. **Payment truth**
   Judge scans QR, pays → PAID after webhook; replay → ignored

6. **Compliance**
   8:30 PM nudge blocked; tamper → chain breaks

7. **Visibility**
   Dashboard, sentiment map, leaderboard with proofs

8. **Close**
   One case. One trail. Full visibility.

RecoverIQ

One case. One trail. Full visibility.

Thank you. We'd love to hear what would make this deployable at your bank tomorrow.

```
Run it:  python run.py  →  http://localhost:8000  ·  demo password Demo@1234
```